

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Smart Lender is a machine learning-powered web application that predicts the creditworthiness of loan applicants, enabling banks and financial institutions to make faster, data-driven loan approval decisions. By evaluating Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost classification models, the system identifies the best-performing model and deploys it for real-time prediction, reducing non-performing assets and improving the overall credit evaluation process.

Project Overview	
Objective	The primary objective is to revolutionize the loan approval process by implementing machine learning techniques (Decision Tree, Random Forest, KNN, and XGBoost) to deliver faster and more accurate creditworthiness assessments.
Scope	The project comprehensively assesses applicant data such as income, employment, credit history, and loan details, and builds a deployable web application that predicts loan approval in real time, reducing manual review effort.
Problem Statement	
Description	Manual and inconsistent credit evaluation processes lead to slow loan approvals, higher rates of non-performing assets, and reduced customer satisfaction.
Impact	An automated, accurate prediction system improves operational efficiency, reduces credit risk, speeds up decision-making, and enhances customer and organizational outcomes.
Proposed Solution	
Approach	Train and evaluate multiple classification algorithms (Decision Tree, Random Forest, KNN, XGBoost) on applicant data, select the best-performing model, and integrate it into a Flask web application for real-time creditworthiness prediction.
Key Features	- Machine learning-based credit assessment model (best model: XGBoost, 94.7% training / 81.1% testing accuracy). - Real-time decision-making for quicker loan approvals via a Flask web interface. - Designed for deployment on IBM Cloud for scalable, accessible predictions.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, xgboost
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv